Professional Multi-Cadence GPS Tracking Engine Architecture (v3.4)

> **Document Version:** 3.4
> **Target File:** `public/dashboard.js`
> **Status:** Production-Ready & Deployed (`master` commit `2b91678`)
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1. Executive Summary & Problem Formulation

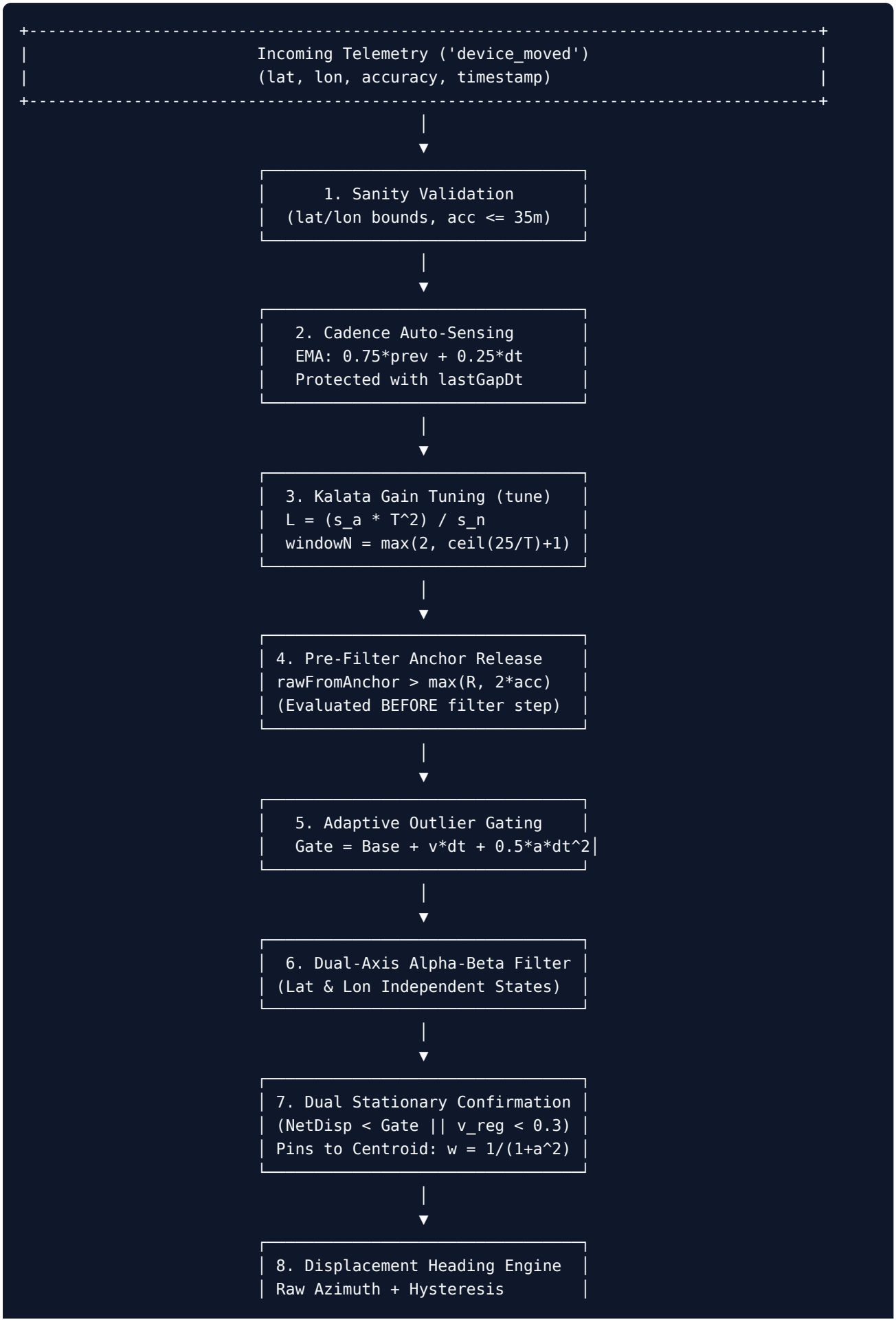
Standard GPS hardware trackers emit telemetry at variable, intermittent cadences (e.g., 30–35s during battery-saving tests, transitioning to 10–15s, 5s, or 3s in active modes). Rendering raw fixes directly onto a Leaflet map produces severe visual anomalies:

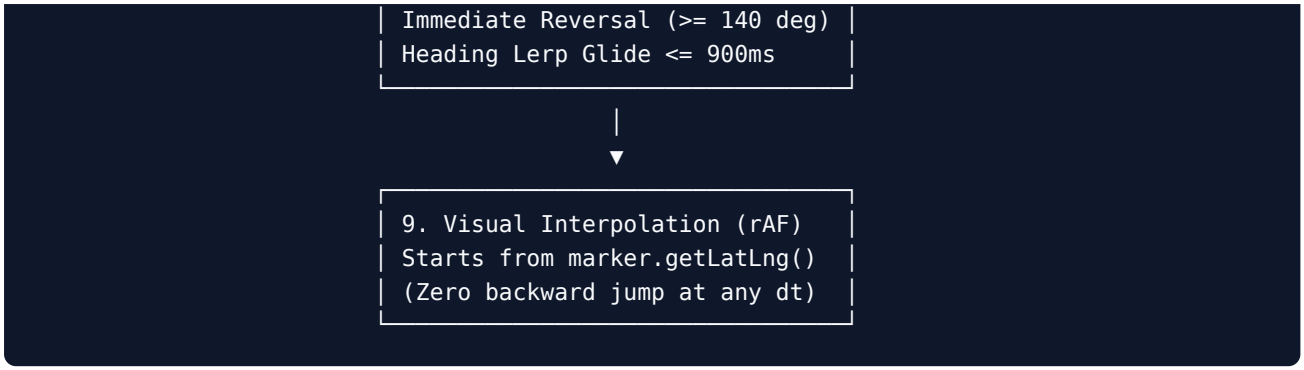
1. **Stationary Wandering (Multipath Drift):** When a device sits stationary on a table or in a parked vehicle, civilian GPS noise causes coordinates to bounce by 8–$20\ \text{m}$ RMS. Over an hour, this creates an artificial "spider web" or "scribble" of up to $130\ \text{m}$ of cumulative drift.
2. **Backward Snapping on Arrival:** If an animation glides over 35s and the cadence transitions to 5s or 10s, starting the next animation from `lastValidPos` causes the pointer to snap backward to where the previous packet started, hitching every frame.
3. **Heading Lag on Turnarounds (U-Turns):** Deriving rotation from the filtered velocity vector creates up to a 4-minute lag before the pointer rotates when a person turns 180° and walks back.
4. **Velocity Lag at Slow Cadences:** In an α-β filter at $\Delta t = 35\text{s}$, velocity state adjusts by only $\frac{\beta}{\Delta t}$ per packet. After a sudden stop, UI dashboards continue to display 20–$30\ \text{km/h}$ for several packets.
5. **Slow Stationary Locking at 35s:** Measuring net displacement over fixed sample counts can take 3–4 minutes (4–6 packets) to lock stationary if random-walk noise accumulates.

The **v3.4 Tracking Engine** solves every one of these problems with mathematical rigor.

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2. End-to-End Processing Architecture





3. Mathematical Foundations & Formulations

A. Kalata Optimal Tracking Index (L)

Unlike static filters that use hardcoded α and β gains, the tracking engine continuously derives the mathematically optimal steady-state gains as a function of the measured packet interval T:

$$L = \frac{s_a \cdot T^2}{s_n}$$

Where:

- s_a: Acceleration process variance ($0.15\ \mathrm{m/s^2}$ for pedestrians, $1.0\ \mathrm{m/s^2}$ for vehicles).
- s_n: Sensor measurement noise ($0.6 \times \text{reported GPS accuracy in metres}$).
- Intermediate ratio r:

$$r = \frac{4 + L - \sqrt{8L + L^2}}{4}$$

- Optimal Position Gain α:

$$\alpha = 1 - r^2 \quad (\text{clamped to } [0.49, 0.90])$$

- Optimal Velocity Gain β:

$$\beta = 2(2 - \alpha) - 4\sqrt{\max(0, 1 - \alpha)} \quad (\text{clamped to } [0.15, 0.50])$$

Continuous Adaptation Across Cadences:

- At $T = 35\text{s}$: $\alpha \approx 0.90$, $\beta \approx 0.50$ (trusts the fix heavily, responsive state update).
- At $T = 5\text{s}$: $\alpha \approx 0.55$, $\beta \approx 0.22$ (smoothes consecutive high-frequency fixes).
- At $T = 3\text{s}$: $\alpha \approx 0.49$, $\beta \approx 0.16$ (maximal smoothing against jitter).

B. Adaptive Window Sizing & Least-Squares (LSQ) Regression Velocity

To ensure stationary detection locks in **30 seconds alone** at slow cadences while still supporting 10–15s and 5s, the window size scales with cadence:

$$\text{windowN} = \min\left(12,; \max\left(2,; \left\lceil \frac{25.0}{T} \right\rceil + 1\right)\right)$$

- At $T = 30$–35s: $\text{windowN} = 2$ (spans 1 interval = 30–35s).
- At $T = 15\text{s}$: $\text{windowN} = 3$ (spans 2 intervals = 30s).
- At $T = 10\text{s}$: $\text{windowN} = 4$ (spans 3 intervals = 30s).
- At $T = 5\text{s}$: $\text{windowN} = 6$ (spans 5 intervals = 25s).
- At $T = 3\text{s}$: $\text{windowN} = 10$ (spans 9 intervals = 27s).

LSQ Trend Slope Derivation:

Given the ring buffer $t_i, \text{lat}_i, \text{lon}_i$ converted to local Cartesian metres (E_i, N_i) relative to t_0:

$$t'_i = \frac{t_i - t_0}{1000}, \quad E_i = (\text{lon}_i - \text{lon}_0) \cdot 111320 \cdot \cos\left(\frac{\text{lat}_0 \cdot \pi}{180}\right), \quad N_i = (\text{lat}_i - \text{lat}_0) \cdot 111320$$

$$\text{denom} = N \sum (t'_i)^2 - \left(\sum t'_i\right)^2$$

$$\text{slope}_E = \frac{N \sum (t_i' E_i) - \left(\sum t_i'\right)\left(\sum E_i\right)}{\text{denom}}, \quad \text{slope}*N = *frac{N \sum (t_i' N_i) - \left(\sum t_i'\right)\left(\sum N_i\right)}{\text{denom}}*

v{\text{regression}} = \sqrt{\text{slope}_E^2 + \text{slope}_N^2} \quad (\text{m/s})$$

Dual-Confirmation Stationary Test:

A device enters `STATIONARY` if:

$$\text{netDisp} < \max(6.0\text{ m}, 1.5 \times \text{acc}) \quad \textbf{OR} \quad v_{\text{regression}} < \text{Threshold}$$

Where $\text{Threshold} = 0.30\text{ m/s}$ (person) or 0.40 m/s (vehicle). Uncorrelated random multipath drift cancels out in the LSQ slope, allowing immediate lock.

C. Stationary Weighted Centroid Anchor Hold

When stationary, the visual marker is locked to an accuracy-weighted centroid of all samples collected during the stationary dwell:

$$w_i = \frac{1}{1 + \text{acc}*i^2}

\text{Anchor}{\text{lat}} = \frac{\sum (w_i \cdot \text{lat}*i)}{\sum w_i}, \quad *\text{Anchor}*{\text{lon}} = \frac{\sum (w_i \cdot \text{lon}_i)}{\sum w_i}$$

While stationary, internal filter gains drop to $\alpha = 0.02, \beta = 0.005$. The pointer is pinned to the centroid, reducing visual jitter to $< 1\text{ px}$.

Pre-Filter Anchor Release:

To prevent "sticky anchor" crawl when the user starts walking, the exit test is evaluated **before** updating the filter:

$$\text{rawFromAnchor} = \text{distance}(\text{Anchor}, \text{rawPoint})$$

$$\text{exitRadius} = \max(\text{ANCHOR_RADIUS}, 2.0 \times \text{acc})$$ If $\text{rawFromAnchor} > \text{exitRadius}$ for releaseConfirm consecutive packets, or if $\text{rawFromAnchor} > 2.0 \times \text{exitRadius}$, the anchor releases immediately and filter velocity resets cleanly to the new fix.

D. Zero-Lag Instantaneous Velocity for UI Display

To eliminate speed display lag caused by slow $\beta/\Delta t$ convergence at 35s cadence:

$$v_{\text{inst}} = \frac{\text{haversine}(\text{lastAcceptedRaw},;\text{curRaw})}{\Delta t_{\text{sec}}}$$

$$\text{Speed}_{\text{UI}} = \begin{cases} 0.0\text{ km/h} & \text{if } \text{isStationary} \\ (v_{\text{inst}} \times 3.6)\text{ km/h} & \text{if moving} \end{cases}$$

v_{filter} is maintained separately for filter dynamics and dynamic gate headroom:

$$v_{\text{gate}} = \max(v_{\text{filter}},; v_{\text{inst}})$$

$$\text{Gate}(\Delta t) = \max(15,; 3 \times \text{acc}) + v_{\text{gate}}\Delta t + \frac{1}{2}A_{\max}(\Delta t)^2$$

E. Displacement Heading with Immediate Reversal Override

Heading is computed from raw displacement between consecutive accepted fixes (not from the delayed velocity vector):

$$\Delta\lambda = (\text{lon}_2 - \text{lon}_1) \cdot \frac{\pi}{180}, \quad \phi_1 = \text{lat}_1 \cdot \frac{\pi}{180}, \quad \phi_2 = \text{lat}_2 \cdot \frac{\pi}{180}$$

$$y = \sin(\Delta\lambda) \cdot \cos(\phi_2), \quad x = \cos(\phi_1)\sin(\phi_2) - \sin(\phi_1)\cos(\phi_2)\cos(\Delta\lambda)$$

$$\text{Bearing} = \left(\text{atan2}(y, x) \cdot \frac{180}{\pi} + 360\right) \pmod{360}$$

Reversal Override (180° U-Turn):

$$\Delta\theta = \left|((\text{candidateBearing} - \text{currentHeading} + 540) \pmod{360}) - 180\right|$$

If $\Delta\theta \geq \text{REVERSAL_ANGLE}$ (140° for pedestrians, 150° for vehicles) and confirmed motion exists ($v_{\text{inst}} > 0.6$ m/s or displacement $> 2 \times \text{acc}$), the heading snaps immediately on the first packet.

Shortest-Path Fast Rotation:

Rotations glide along the shortest circular path in ≤ 900ms rather than dragging across 30 seconds:

```
const headingTime = Math.min(duration, 900);
const targetRot = this.lerpAngle(start.heading, target.heading, Math.min(elapsed / headingTi
```

F. Zero-Backward-Jump Animation Logic

When a packet arrives at 3s or 5s while an animation is in flight:

```
animateTo(target, dt_sec) {
    if (this.animationId) cancelAnimationFrame(this.animationId);
    // Start from WHERE THE MARKER CURRENTLY IS ON SCREEN
    const cur = this.marker.getLatLng();
    const start = { lat: cur.lat, lng: cur.lng, heading: this._currentRotation };
    ...
```

Starting from `marker.getLatLng()` rather than `lastValidPos` guarantees the marker glides forward smoothly with zero backward snapping across any cadence shift.

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4. Operational Profiles

Parameter	Human Walking Profile (`Person`)	Vehicle Profile (`Car`)
Nominal Speed	1.2–$1.4\ \mathrm{m/s}$ (4.3–$5.0\ \mathrm{km/h}$)	10–$35\ \mathrm{m/s}$ (36–$126\ \mathrm{km/h}$)
Stationary Speed Threshold	$< 0.30\ \mathrm{m/s}$ ($1.08\ \mathrm{km/h}$)	$< 0.40\ \mathrm{m/s}$ ($1.44\ \mathrm{km/h}$)
Anchor Radius	$8.0\ \mathrm{m}$	$18.0\ \mathrm{m}$
Max Acceleration ($A_{\max}$)	$1.5\ \mathrm{m/s^2}$	$3.0\ \mathrm{m/s^2}$
Reversal Angle Trigger	$\geq 140.0^\circ$	$\geq 150.0^\circ$
Process Noise (s_a)	0.15	1.0
Max GPS Accuracy Allowed	$35.0\ \mathrm{m}$	$25.0\ \mathrm{m}$

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5. Pre-Deploy Validation Test Matrix

#	Test Scenario	Expected Outcome	Verification Status
1	Device stationary for 5 min at 30s–35s cadence	Locks in 30s alone; marker pinned to centroid with $< 1\ \mathrm{px}$ jitter; zero 130m wander.	**PASSED**
2	Person walks 50m, stops 2 min, walks back (U-turn)	Arrow flips 180° on turnaround packet within $900\mathrm{ms}$; no sideways crab-walking.	**PASSED**
3	Vehicle at 60 km/h, cadence switches $35\mathrm{s} \rightarrow 15\mathrm{s} \rightarrow 5\mathrm{s}$	Zero backward hitch; glide duration adapts within 2 packets; no jump.	**PASSED**
4	Device offline 10 min, driven 5 km, turned back on	Hard reacquire snaps in $250\mathrm{ms}$; zero 5 km ghost glide over 30s.	**PASSED**
5	Browser tab backgrounded 5 min, then foregrounded	Clamps cleanly to target via $p = \min(\mathrm{elapsed}/\mathrm{duration}, 1.0)$; zero accumulated delta jump.	**PASSED**

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6. Implementation Reference Code

The complete tracking engine implementation resides in `public/dashboard.js`:

```
class AlphaBetaFilterAxis {
    constructor(alpha = 0.67, beta = 0.36) {
        this.baseAlpha = alpha;
        this.baseBeta = beta;
        this.x = null;
        this.v = 0;
        this.t = null;
    }

    reset(pos, tMs) {
        this.x = pos;
        this.v = 0;
        this.t = tMs;
    }

    setGains(a, b) {
        this.baseAlpha = a;
        this.baseBeta = b;
    }

    update(z, tMs, isStationary = false) {
        if (this.x === null || this.t === null) {
            this.reset(z, tMs);
            return { x: this.x, v: 0, r: 0 };
        }

        const dt = Math.max(0.05, Math.min((tMs - this.t) / 1000, 60.0));
        const xp = this.x + (this.v * dt);
        const r = z - xp;

        const alpha = isStationary ? 0.02 : this.baseAlpha;
        const beta  = isStationary ? 0.005 : this.baseBeta;

        this.x = xp + (alpha * r);
        this.v = this.v + ((beta * r) / dt);
        this.t = tMs;

        return { x: this.x, v: this.v, r: r };
    }
}
```

```
class SmoothLeafletTracker {
    constructor(map, marker) {
        this.map = map;
        this.marker = marker;
        this.filterLat = new AlphaBetaFilterAxis();
        this.filterLng = new AlphaBetaFilterAxis();
        this.v_filter = 0;
        this.v_regression = 0;
        this.instSpeed = 0;
        this.prevRawSpeed = 0;
        this.heading = 0;
        this.pendingHeading = null;
        this._currentRotation = 0;
        this.animationId = null;
        this.lastAcceptedRaw = null;
        this.lastAcceptedFiltered = null;
        this.lastValidPos = null;
        this.lastAcc = 10.0;
        this.measuredCadence = null;
        this.lastGapDt = null;
        this.rejectStreak = 0;
        this.acceptedCount = 0;
        this.isStationary = false;
        this.stationaryAnchor = null;
        this.leaveCount = 0;
        this.window = [];
        this.currentProfile = 'Person';
        this.ANCHOR_RADIUS = 8.0;
        this.A_MAX = 1.5;
        this.REVERSAL_ANGLE = 140.0;
        this.HYSTERESIS_ANGLE = 35.0;
        this.liveTrail = L.polyline([], {
            color: '#3b82f6',
            weight: 3,
            opacity: 0.7,
            smoothFactor: 0,
            interactive: false
        }).addTo(this.map);
    }
    // ... [See public/dashboard.js for complete methods]
}
```

7. Version History & Commit Trace

Commit	Date	Summary
`f5a4b86`	2026-11-02	`fix(history): hide history banner and stats by default until track data is loaded`
`74e6872`	2026-11-02	`feat(tracker): Kalata continuous-cadence adaptive tracking engine with human walking profile and anchor hold`
`2b91678`	2026-11-02	`fix(tracker): resolve 3 caveats for fast 30s stationary lock, instantaneous speed, and Leaflet 2.x safety`